

# 1 Notation

We model four events each year: growth, colonization, fizzle and end-of-wave impact.

Let  $(k, t)$  be the time index, where  $k$  represents the year index and  $t$  represents the within-year event index:  $t = 0$  for the time at the beginning of the year,  $t = 1$  for the time after growth,  $t = 2$  for the time after colonization,  $t = 3$  for the time after fizzle, and  $t = 4$  for the time at the end of the year (after the wave). We interpret  $(k, 0)$  as the same time as  $(k - 1, 4)$ .

Assume there are  $n$  patches and we label them with  $p_i$  ( $i = 1, \dots, n$ ). For each patch we use  $(N_i^{(k,t)}, I_i^{(k,t)})$  to describe its state.  $N_i^{(k,t)}$  represents population size in patch  $p_i$  at time  $(k, t)$ , and

$$I_i^{(k,t)} \in \{0, 1, 2, \dots\}$$

is the number of infected hosts in patch  $p_i$  at time  $(k, t)$ .

We define the (derived) infection indicator

$$\mathbb{1}_i^{(k,t)} = \mathbb{1}\{I_i^{(k,t)} \geq 1\}.$$

Let  $U \sim \text{Uniform}(0, 1)$  be a random variable uniformly distributed over  $(0, 1)$ , and we define

$$\mathbb{1}_{U \leq P} = \begin{cases} 1, & \text{if } U \leq P, \\ 0, & \text{otherwise.} \end{cases}$$

# 2 Parameters

Host:

- $r$ : growth rate
- $K$ : carrying capacity of a patch (all patches are of equal area)
- $c \in [0, 1]$ : probability that an individual host moves to another patch during colonization

Epidemic:

- $\mathcal{R}_0$ : basic reproduction number
- $P_f = 1/\mathcal{R}_0$ : fizzle probability for a *single* introduced infection
- $z = z(\mathcal{R}_0)$ : final size fraction of a non-fizzling within-patch epidemic wave
- $D$ : disease-induced mortality fraction among infected hosts (assume  $D = 1$ )

Propagule (infectious fleas):

- $\nu$ : scaling constant linking the previous year's epidemic size in a patch to number of infectious propagules
- $\eta$ : scaling constant linking host movement probability  $c$  to propagule movement probability  $g$

- We define the propagule movement probability (between patches) as

$$g = \min\{1, \eta c\}, \quad (1)$$

to ensure  $g \leq 1$

Infection:

- $\beta \geq 0$ : mean number of effective bites per propagule during the colonization season
- $p \in [0, 1]$ : probability that a single effective bite infects the bitten host
- $\kappa = \beta p \nu$

**Propagule assumption** We assume free fleas do not survive beyond the current season, so all propagules relevant for year  $k$  are generated by the year  $k - 1$  wave.

### 3 Growth

We assume that reproduction of rats follows logistic growth:

$$N_i^{(k,1)} = N_i^{(k,0)} + r N_i^{(k,0)} \left( 1 - \frac{N_i^{(k,0)}}{K} \right). \quad (2)$$

Assume infection counts do not change during growth:

$$I_i^{(k,1)} = I_i^{(k,0)} = 0. \quad (3)$$

## 4 Colonization and Fizzle

### 4.1 Population size

We assume that each individual rat moves with probability  $c$  and lands in a random patch. We ignore distances and assume that all patches are equally available to a moving rat:

$$N_i^{(k,2)} = (1 - c) N_i^{(k,1)} + \frac{c \sum_{j=1}^n N_j^{(k,1)}}{n}. \quad (4)$$

The first term is population staying in the patch and the second term is population moving into the patch.

### 4.2 Propagules generated by the previous year

Let  $\mathbb{1}_j^{(k-1,3)} = \mathbb{1}\{I_j^{(k-1,3)} \geq 1\}$  indicate whether patch  $p_j$  is infected after the fizzle step in year  $k - 1$ .

The number of infected and dead hosts produced by the wave in patch  $j$  in year  $k - 1$  is

$$S_j^{(k-1)} = z D N_j^{(k-1,3)} \mathbb{1}_j^{(k-1,3)}. \quad (5)$$

**Propagule production** We assume the number of infectious propagules generated by patch  $j$  for year  $k$  is

$$F_j^{(k)} \mid S_j^{(k-1)} \sim \text{Poisson}\left(\nu S_j^{(k-1)}\right). \quad (6)$$

Let  $R_j^{(k)}$  be the number of propagules generated by patch  $j$  that remain in the same patch. Conditional on  $F_j^{(k)}$ :

By Poisson thinning of (6):

$$R_j^{(k)} \mid S_j^{(k-1)} \sim \text{Poisson}\left((1-g)\nu S_j^{(k-1)}\right). \quad (7)$$

Let  $E_j^{(k)}$  be the number of propagules generated by patch  $j$  that are exported to other patches:

$$E_j^{(k)} \mid S_j^{(k-1)} \sim \text{Poisson}\left(g\nu S_j^{(k-1)}\right). \quad (8)$$

Exported propagules are assumed to land uniformly at random among the  $n$  patches. The number of propagules exported from source patch  $j$  that arrive in target patch  $i$  is

$$Y_{j \rightarrow i}^{(k)} \mid S_j^{(k-1)} \sim \text{Poisson}\left(\frac{g\nu}{n} S_j^{(k-1)}\right). \quad (9)$$

Summing over sources gives the total number of imported propagules in patch  $i$ :

$$Y_{i,\text{import}}^{(k)} = \sum_{j=1}^n Y_{j \rightarrow i}^{(k)}. \quad (10)$$

**Total propagules present in patch  $i$ .** The total number of propagules present in patch  $i$  at colonization is

$$G_i^{(k)} = R_i^{(k)} + Y_{i,\text{import}}^{(k)}. \quad (11)$$

### 4.3 newly infected hosts

We now derive the per-host infection probability and the distribution of the number of newly infected hosts created during colonization.

**Per-host infection probability.** Fix a particular host in patch  $i$ . Given  $G_i^{(k)}$  propagules, assume: (i) each propagule makes a  $\text{Poisson}(\beta)$  number of effective bites; (ii) each bite chooses a host uniformly among the  $N_i^{(k,2)}$  hosts in the patch; (iii) each bite infects with probability  $p$ , independently.

Then the number of *successful infection attempts* on the focal host is Poisson with mean

$$\frac{\beta p G_i^{(k)}}{N_i^{(k,2)}}.$$

Therefore the probability that the focal host becomes infected is

$$\pi_i^{(k)} = 1 - \exp\left(-\frac{\beta p G_i^{(k)}}{N_i^{(k,2)}}\right) \approx \frac{\beta p G_i^{(k)}}{N_i^{(k,2)}}. \quad (12)$$

**Number of newly infected hosts** Given  $\pi_i^{(k)}$ , the number of hosts infected is

$$X_i^{(k)} \mid \left(G_i^{(k)}, N_i^{(k,2)}\right) = \text{Binomial}\left(N_i^{(k,2)}, \pi_i^{(k)}\right).$$

When  $\pi_i^{(k)} \ll 1$ , we have

$$\begin{aligned} X_i^{(k)} \mid \left(G_i^{(k)}, N_i^{(k,2)}\right) &\approx \text{Poisson}\left(N_i^{(k,2)} \pi_i^{(k)}\right) \\ &= \text{Poisson}\left(\beta p G_i^{(k)}\right) \\ &= \text{Poisson}\left(\beta p \nu z D \left[ (1-g) N_i^{(k-1,3)} \mathbb{1}_i^{(k-1,3)} + \frac{g}{n} \sum_{j=1}^n N_j^{(k-1,3)} \mathbb{1}_j^{(k-1,3)} \right]\right) \end{aligned} \quad (13)$$

Let  $\kappa = \beta p \nu$

**Update infected count after colonization.** We define

$$I_i^{(k,2)} = I_i^{(k,1)} + X_i^{(k)} = X_i^{(k)}. \quad (14)$$

## 5 Fizzle

Let  $P_f = 1/\mathcal{R}_0$  be the probability that a single introduction fails to establish. If  $X$  introductions occur independently, then the probability that all of them fizzle is  $P_f^X$ . Therefore,

$$P(\text{establish}) = 1 - \mathbb{E}[P_f^X].$$

Since  $X \sim \text{Poisson}(\lambda)$ ,

$$\mathbb{E}[P_f^X] = \sum_{x=0}^{\infty} P_f^x \frac{e^{-\lambda} \lambda^x}{x!} = e^{-\lambda} \sum_{x=0}^{\infty} \frac{(\lambda P_f)^x}{x!} = e^{-\lambda} e^{\lambda P_f} = e^{-\lambda(1-P_f)}.$$

Therefore

$$P(\text{establish}) = 1 - e^{-\lambda(1-P_f)}$$

where

$$\lambda = \kappa z D \left[ (1-g) N_i^{(k-1,3)} \mathbb{1}_i^{(k-1,3)} + \frac{g}{n} \sum_{j=1}^n N_j^{(k-1,3)} \mathbb{1}_j^{(k-1,3)} \right]$$

Population size does not change:

$$N_i^{(k,3)} = N_i^{(k,2)}. \quad (15)$$

## 6 Major epidemic

In patches that experience major epidemic, the population is reduced by the epidemic wave:

$$N_i^{(k,4)} = \left(1 - z D \mathbb{1}_i^{(k,3)}\right) N_i^{(k,3)}. \quad (16)$$

We assume infected rats die quickly and no infected rats remain at the end of a wave:

$$I_i^{(k,4)} = 0 \quad \text{for all } i. \quad (17)$$

## 7 Year-to-year update

$$N_i^{(k+1,0)} = N_i^{(k,4)}, \quad I_i^{(k+1,0)} = I_i^{(k,4)} = 0. \quad (18)$$